AE 5335 Midterm

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I. Methodology

For this assignment a code was developed to integrate the quasi-1D flow equations using the callorically perfect gas assumption. The equations of state are:

$$\frac{d\rho}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0 \tag{1}$$

$$\frac{dP}{\rho} + udu = \frac{-\tau_w c dx}{\rho A} + \eta \delta w_{shaft} \tag{2}$$

$$C_p dT + udu = \delta q + \delta w_{shaft} \tag{3}$$

$$\frac{dP}{P} = \frac{d\rho}{\rho} + \frac{dT}{T} \tag{4}$$

In this current form we are left with a system of 4 equations with 4 unknowns which can be solved for using a root solving algorithm. However, since root solvers themselves can be computationally expensive and introduce numerical additional numerical errors an additional step was made. The system of equations shown above was reduced algebraically to a single equation with one unknown. The variable that was solved for was du due to its presence in the continuity, momentum, and energy equations.

$$du = -\frac{\frac{\rho}{P}\left(\eta dw - \frac{\tau_w c dx}{\rho A}\right) - \frac{dq+dw}{CpT} + \frac{dA}{A}}{\frac{1}{u} - \frac{u}{RT} + \frac{u}{CpT}} \tag{5}$$

$$\left(\frac{x}{L}\right)_{T_w=T_s} = 1 + \frac{1}{\pi}\tan^{-1}\frac{-\pi D}{2C_f L} \rightarrow x_{T_w=T_s} = \boxed{0.9031078\ m} \tag{6}$$

Use this equation to find ΔT_m:

$$T_s = T_{t_1} + \frac{2C_F L}{\pi D}\Delta T_m \left(1 + \sqrt{1 + \left(\frac{\pi D}{2C_f L}\right)^2}\right) \rightarrow \Delta T_m = 460.152\ K \tag{7}$$

We can then use this equation to find the total temperature at station 2:

$$T_{t_2} - T_{t_1} = \frac{4C_F L}{\pi D}\Delta T_m \rightarrow T_{t_2} = 3129.420\ K \tag{8}$$

Since we know the change in temperature through the reactor we can calculate the heat generated using:

$$\dot{Q} = \dot{m} c_p \left(T_{t_2} - T_{t_1}\right) = \boxed{21.147\ MW} \tag{9}$$

Where:

$$c_p = \frac{\gamma}{\gamma - 1} R = 14437.5 \frac{J}{kgK} \tag{10}$$

The following equation, along with a root solver (scipy.optimize.fsolve), can be used to find the mach number at station 2:

$$\sqrt{\frac{T_{t_2}}{T_{t_1}}} = \frac{M_2}{M_1}\left[\frac{1 + \gamma M_1^2\left[1 - \frac{C_f L}{D}\right]}{1 + \gamma M_2^2\left[1 + \frac{C_f L}{D}\right]}\right]\left[\frac{1 + \frac{\gamma-1}{2}M_2^2}{1 + \frac{\gamma-1}{2}M_1^2}\right]^{\frac{1}{2}} \rightarrow M_2 = 0.138137 \tag{11}$$

The static pressure ratio is:

$$\frac{P_2}{P_1} = \left[\frac{1 + \gamma M_1^2\left[1 - \frac{C_f L}{D}\right]}{1 + \gamma M_2^2\left[1 + \frac{C_f L}{D}\right]}\right] = 0.85751166 \tag{12}$$

Then we can find the pressure at location 2:

$$\frac{P_{t_2}}{P_{t_1}} = \frac{P_2}{P_1}\left[\frac{1+\frac{\gamma-1}{2}M_2^2}{1+\frac{\gamma-1}{2}M_1^2}\right]^{\frac{\gamma}{\gamma-1}} = 0.86847317 \rightarrow P_{t_2} = 173.695\ kPa \tag{13}$$

This total pressure and temperature is also the total pressure and temperature at the exit of the nozzle given that we have an isentropic nozzle.

Because the nozzle is isentropic we can use the following equation to find the Mach number at the nozzle exit:

$$\dot{m}_{H_2} = \frac{P_{t_e}A_e}{\sqrt{T_{t_e}}}\sqrt{\frac{\gamma}{R}}M_e(1+\frac{\gamma-1}{2}M_e^2)^{\frac{-(\gamma+1)}{2(\gamma-1)}} \rightarrow M_e = 6.99997 \approx 7.0 \tag{14}$$

Using the Mach number, total temperature, and total pressure we can find the temperature and pressure at the exit.

Exit temperature is:

$$T_e = \frac{T_{te}}{1+\frac{\gamma-1}{2}M_e^2} = 289.763\ K \tag{15}$$

Exit pressure is:

$$P_e = \frac{P_{te}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = 41.958\ Pa \tag{16}$$

The velocity at the exit is:

$$u_e = M_e \sqrt{\gamma R T_e} = 9055.114\ m/s \tag{17}$$

Because the control volume is drawn around the propellant tanks there is only mass flow leaving the system. Because of this we calculate the thrust using only the exit velocity and pressure. Also since the external pressure should be close to zero that term cancels out.

The thrust produced by the engine is:

$$F = \dot{m}_p u_e + P_e A_e = \boxed{4593.557\ N} \tag{18}$$

Then the specific impulse can be found with:

$$I_{sp} = \frac{F}{\dot{m}_p g_0} = \boxed{936.505\ sec} \tag{19}$$

The total temperature of the gas and the wall with respect to their axial location can be plotted using the following equations:

$$T_t = T_{t_1} + \frac{2C_f}{D} \Delta T_m \frac{L}{\pi} \left[1 - \cos \frac{\pi x}{L}\right] \tag{20}$$

$$T_w = T_{t_1} + \frac{2C_f}{D} \Delta T_m \frac{L}{\pi} \left[1 - \cos \frac{\pi x}{L}\right] + \Delta T_m \sin \frac{\pi x}{L} \tag{21}$$

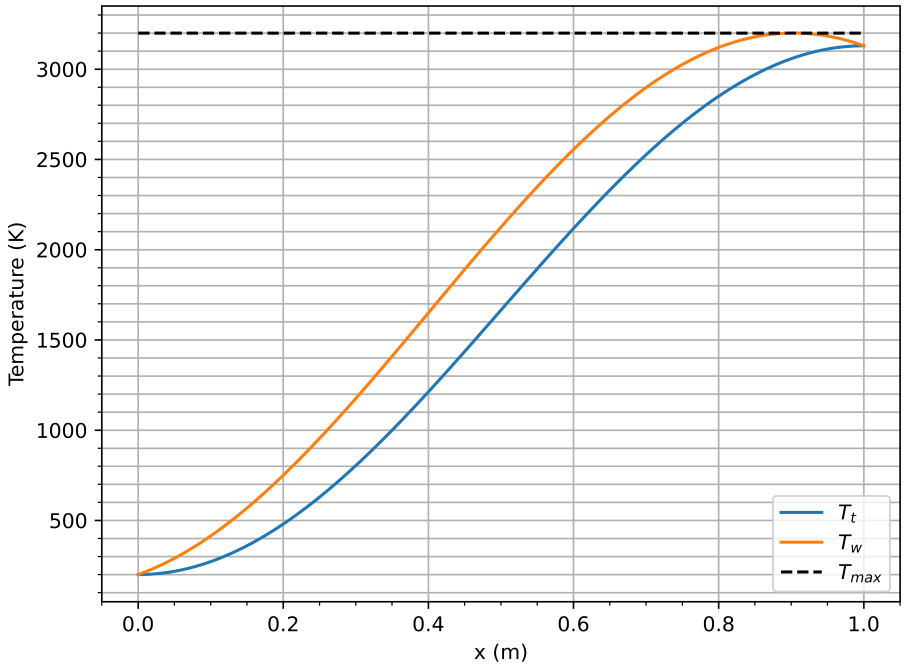


Fig. 1 Gas total temperature and wall temperature along the axial direction